From Retrospective Value to Prospective Design

Valuing Forecast Improvements Under a Nonstationary Climate

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Roadmap

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I: What We Know

The Retrospective Result

We quantified the social value of augmented observations (AO) for eight landfalling hurricanes (2017–2020), combining data-denial experiments (Ditchek et al. 2023) with a structural cost model (Molina et al. 2024).

Three key estimates:

1. **Loss curvature:** $\widehat{\beta}_2^w = 5.16$ million per unit increase in squared demeaned wind error, for hurricane-force counties.
2. **Precision gain from AO:** $\widehat{\delta} = 0.26$ m/s reduction in forecast standard deviation at the 72-hour lead time.
3. **Per-county value:** $\approx$ \$24 million per affected county per storm.

Benefit-to-cost ratio $\approx$ 50:1 against flight operations.

The Valuation Formula

Hurricane costs depend on two features of the forecast environment: the *level* of the storm (mean intensity μ) and the *precision* of the forecast (error variance σ^2). The expected cost can be written as:

$$\mathbb{E}[C] = f(\mu) + \hat{\beta}_2^w \cdot \sigma^2$$

where $f(\mu)$ captures the direct cost of storm intensity and $\hat{\beta}_2^w \cdot \sigma^2$ captures the cost of forecast uncertainty. The social value of a precision improvement that reduces error variance from σ_0^2 to σ_1^2 is:

$$V = \hat{\beta}_2^w \left(\sigma_0^2 - \sigma_1^2\right)$$

Key property: Because hurricane damages are *convex* in wind speed, both the level μ and the scale σ^2 matter. A higher μ increases the baseline cost and can amplify the marginal cost of errors. A higher σ^2 raises the baseline from which improvements must operate.

The Stationarity Assumption

All retrospective estimates draw from the 2005–2022 climate.

The valuation formula treats four objects as fixed:

1. $\widehat{\beta}_2^w$: the loss curvature linking forecast errors to costs.
2. μ: the location of the intensity distribution (mean wind speed for affected counties).
3. σ_0^2: the baseline forecast error variance (status quo technology).
4. $\Delta\sigma^2$: the variance reduction achievable by a given observational investment.

Under a changing climate, some (or all) of these are stationary. Climate might shift both the *location* μ (higher peak intensities, more Cat 4–5 events) and the *scale* σ^2 (greater forecast uncertainty from expanded atmospheric variability).

II: Why Nonstationarity Matters

Two Channels of Climate Influence

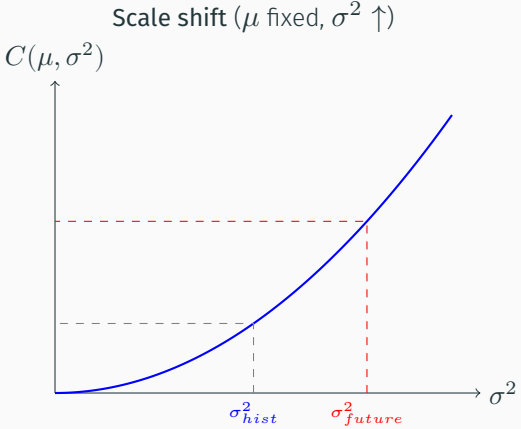
Climate change might change the value of forecast precision through two channels:

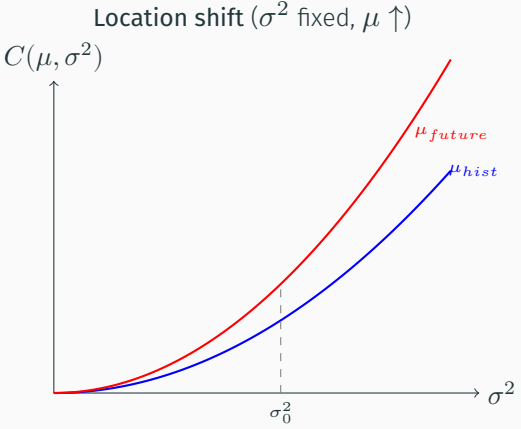
1. Location shift ($\mu \uparrow$). The intensity distribution shifts rightward. Projections might indicate a higher proportion of Category 4–5 events and increased peak wind speeds. More county-storms cross the hurricane-force threshold (> 33 m/s), enlarging the set over which $\hat{\beta}_2^w$ applies. Because damages are convex in wind speed, a higher μ also amplifies the marginal cost of forecast errors at any given σ^2.

2. Scale shift ($\sigma^2 \uparrow$). Greater variability, rapid intensification episodes, and thermodynamic environments increases the irreducible component of forecast error variance. The forecast models must operate on a distributional support that is expanding.

Both channels raise the social value of a given precision improvement $\Delta\sigma^2$.

Convexity: Both Channels Amplify Value





Left: Same $\Delta\sigma^2$, but higher baseline σ^2 means steeper cost curve $\Rightarrow$ larger V. **Right:** Same σ^2 and $\Delta\sigma^2$, but higher μ rotates the cost curve upward $\Rightarrow$ larger V.

III: A Framework for Prospective Valuation

The Forecast Error Model

For location i, lead time s periods ahead from time t:

$$y_{i,t+s} = \hat{g}(X_t) + \epsilon_{i,t+s}, \quad \epsilon_{i,t+s} \sim F_s(0,\ \sigma_s^2)$$

where X_t is the vector of current and lagged atmospheric states and $\hat{g}(\cdot)$ is the forecast model's best prediction. Under climate change, the distribution of outcomes $y_{i,t+s}$ shifts in both location and scale:

- **Location:** The mean of the intensity distribution μ shifts rightward (higher V_{max}, more rapid intensification episodes, etc). This increases the probability of extreme realizations and the baseline cost against which errors are evaluated.
- **Scale:** The conditional forecast error variance σ_s^2 increases as the atmospheric state space expands beyond the historical support on which $\hat{g}(\cdot)$ was estimated.

Both shifts raise the social value of any given precision improvement $\Delta\sigma^2$.

A Prospective Valuation Approach

Define the value of a forecast improvement under climate scenario λ, which determines both the location and scale of the intensity distribution:

$$V(\lambda) = \beta(\mu(\lambda)) \cdot \Delta\sigma^2(\lambda)$$

where:

- $\mu(\lambda)$: the location of the intensity distribution under scenario λ. A higher μ increases baseline damages and can amplify the loss curvature β.
- $\sigma_0^2(\lambda)$: the *status quo* forecast error variance under scenario λ.
- $\Delta\sigma^2(\lambda) = \sigma_0^2(\lambda) - \sigma_1^2(\lambda)$: the variance reduction from the improved technology.

Climate change raises V through both parameters: μ and σ_0^2. These effects compound: the same technology is worth more when storms are both stronger *and* harder to predict.

A Semi-Parametric Simulation

Step 1: Empirical baseline. Estimate the joint distribution of (μ, σ_s^2) from the historical sample of data-denial experiments (e.g., Ditchek et al. 2023), stratified by lead time and intensity category. This is the known object.

Step 2: Climate perturbation. Shift (μ, σ^2) (e.g., under CMIP6 scenarios, SSP2-4.5 vs. SSP5-8.5). The location shift draws on projected changes in TC intensity distributions (e.g., Knutson et al., 2020). The scale shift reflects increased forecast error variance from expanded atmospheric variability.

Step 3: Propagation. For each draw from the perturbed distribution, compute $V(\lambda) = \beta(\mu(\lambda)) \cdot \Delta\sigma^2(\lambda)$ and aggregate across county-storms to obtain scenario-specific total values.

Location channel ($\mu \uparrow$). Use projected rightward shift of the TC intensity distribution to perturb μ.

- Literature suggests a modest increase in mean peak wind (about 3% under 2°C warming) but a substantially heavier right tail, with 10% more Category 4–5 storms.
- A higher μ both enlarges the set of hurricane-force county-storms and steepens the loss function $\beta(\mu)$. **Scale channel ($\sigma^2 \uparrow$).** Scale σ_s^2 by a multiplicative factor $\kappa > 1$ reflecting the increase in irreducible forecast uncertainty.
    - This channel captures the effect of expanding atmospheric variability on model skill and is the more speculative of the two.

An analogous framework applies to track uncertainty. Define location-level exposure as:

$$p_{i,t+s} = \Pr(\text{location } i \text{ within strike zone at } t+s \mid X_t)$$

This can be implemented as:

- A binary indicator: whether the TC centroid passes through location i.
- A kernel-weighted function: exposure that decays with distance from the forecast center, e.g., $K\left(\frac{\|r_i - r_{center}\|}{R}\right)$ for some radius R.

A Simple Illustration

A Simple Illustration: Setup

We compute $V(\mu, \sigma^2) = \beta(\mu) \times \Delta\sigma^2$ over a grid of plausible (μ, σ_0^2) pairs.

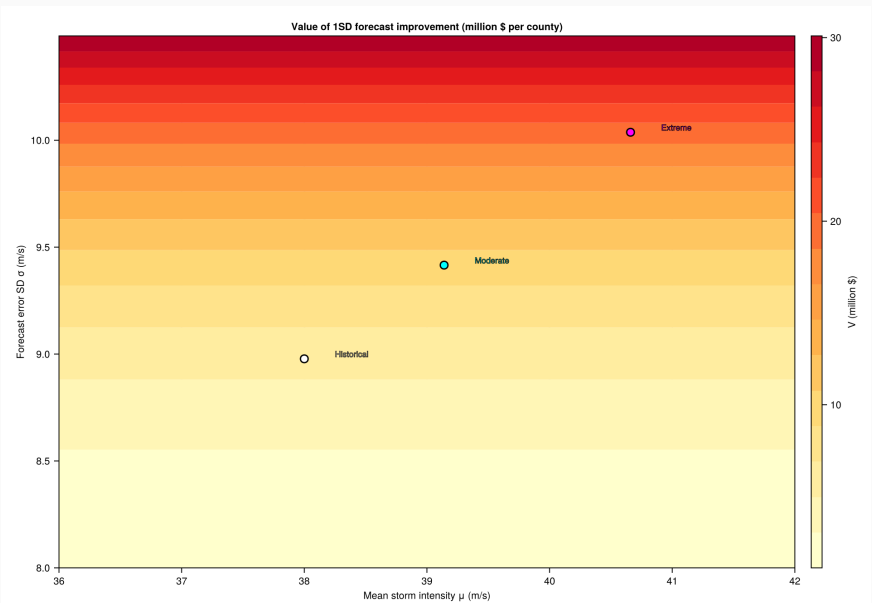
Calibration from the retrospective analysis:

- $\hat{\beta}_2^w = 5.16$ M per (m/s)^2, at a historical mean intensity $\bar{\mu} = 38$ m/s.
- $\Delta\sigma^2 = 1$ (m/s)^2, hypothetical variance reduction from augmented observations.
- $\beta(\mu)$ is increasing in μ above the hurricane-force threshold (33 m/s), reflecting the convexity of damages in wind speed: $\beta(\mu) = \hat{\beta}_2^w \left(\frac{\mu - 33}{\bar{\mu} - 33}\right)^3$.

Tentative climate scenarios shift (μ, σ_0^2) jointly:

- Moderate μ increases 3%, σ^2 scales by $\kappa = 1.10$.
- Extreme μ increases 7%, σ^2 scales by $\kappa = 1.25$.

A Simple Illustration: Value Surface $V(\mu, \sigma^2)$



Open Questions

Open Questions for the Community

1. Calibrating the scale channel. What is the best empirical or model-based strategy for projecting how σ_s^2 scales with changes in the large-scale environment (the scale factor κ)?

2. Structural stability of precision gains. Should we expect the physical improvement from a given observational investment (the $\Delta\sigma^2$) to remain constant as (μ, σ^2) shift? Or does data assimilation performance itself degrade (or improve) in a warmer climate?

3. Multi-hazard extension. The current framework is wind-only. Precipitation and storm surge channels likely amplify the total value. How should we prioritize these extensions, and what data would be needed?

We welcome any and all feedback!